

COMP1886 MACHINE LEARNING
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TASK 1: Identify a scenario and perform a literature study

Scenario Analysis

Germany's equivalent of the UK's Pay As You Earn(PAYE) system is the wage tax, known as 'Lohnsteuer'. It is a withholding mechanism through which income tax is collected from Germans' employment earnings at source. Each time they process payroll, employers calculate, deduct and pay tax to the 'Finanzamt'(German tax office). This is like the Real Time Information(RTI) architecture of the dataset that was analysed in this coursework. 'Lohnsteuer' generated around €248.9 billion in revenue in 2024, which makes Germany's second-largest source of tax revenue after the 'VAT-system'. It accounted for over 26% of total federal and state tax receipts (Destatis, 2025). Germany's wage tax system is one of the most data-rich in the world because it receives a lot of data continuously. Therefore, it is a good candidate for machine learning-based analysis.

The "Einkommenssteuergesetz(ESTG)" is the legal basis of the "Lohnsteuer." Germany has a modern income tax system. This includes a zero-rate band covering earnings, called 'Grundfreibetrag'(basic personal allowance), set at €11,784 in 2024. Whereby taxes increase from 14% to 42% for people who earn between €12,349 and €68,481. Employees who earn more than €277,825 receive the highest rate of 45%. The average single German worker paid 37.4% in taxes (OECD 2024). Whereby the average is predicted with only around 25% accuracy. This places Germany among the countries with the most heavily taxed labour markets in the developed world. The methodological insights derived from German machine learning (ML) research can be directly applied to the mean-pay prediction task conducted herein, as both Germany's 'Lohnsteuer' and the UK's PAYE generate monthly payroll data across various industry sectors.

The "Steuerklassen" framework is a key part of how the German system works. It puts workers into one of six tax brackets, from I to VI. It depends on whether they are married and how many people in their household are working. However, it attracts significant criticism from economists and policymakers. The OECD(2022) demanded changes to Germany's joint taxation system, known as "Ehegattensplitting". The progressive rate structure gives single-earner households the highest tax break and makes it less likely that secondary earners, such as women, will work full-time or increase their hours. Also, the Ifo Institute characterised Germany as one of the few OECD countries whose tax system actively maintains gender inequality in the labour market. Both empirical and moral justifications: According to Destatis(2024), the unadjusted gender pay gap in Germany was 18% in 2023. Fuest, Peichl, and Siegloch's[1] research indicates that the impact of alterations in corporate taxes in Germany is also disproportionately borne by low-skilled, young, and female employees. A distributional pattern that is highly probable to exist in the industry-level pay data examined in this coursework. Therefore, the fact that Germany's tax system fails to address these structural problems is a political concern and a known source of bias in any machine learning model that uses German payroll data.

Another structural problem is that Germany still has trouble accurately predicting "Lohnsteuer" revenues. Göttert and Lehmann(2021)[4] conducted a comprehensive examination of this matter. Their investigation into the inaccuracies in forecasts produced by the 'Arbeitskreis Steuerschätzungen'(the government body responsible for official projections) indicated that these discrepancies are mainly due to incorrect calculations of the fundamental tax base(particularly wages and employment rates) rather than to flawed elasticity assumptions. This is an important point in the current coursework because it means

that the quality of the input dataset is much more important than the model's sophistication. The Mean Pay sheet in the COMP1886 coursework dataset has errors and missing values on purpose to show how this kind of tax base data quality problem can happen. The pre-processing steps in Task 3 are therefore highly important. They directly replicate the data cleaning challenge that undermines real-world tax forecasting in Germany.

Germany's digital tax infrastructure, particularly the ELStAM('Elektronische LohnSteuerAbzugsMerkmale') system, was introduced in 2013. It replaced the old paper-based 'Lohnsteuerkarte' system with a centralised electronic database of employees' tax information. At each payroll cycle, employers retrieve each worker's tax information, including tax class and allowances, from the federal system. While this creates a structured, machine-readable data stream, it also makes the system vulnerable to evasion and reporting inconsistencies. Lahann, Scheid, and Fettke (2019)[2] demonstrate that transactional payroll and accounting datasets are well-suited to machine learning(ML)-based compliance detection. They use decision tree classifiers to identify value-added tax(VAT) violations in German-standard accounting records. Their finding that tree-based models outperform simple statistical methods for categorical and mixed-type tax data is important for the coursework categories, such as task 4.

At the business level, tax avoidance creates a discrepancy between the theoretical obligation to pay taxes and the actual collection of revenue. Using artificial neural network regressions on CompStat data from Germany, the UK, France, Italy and Spain, Delgado et al.(2023)[3]. It revealed significant nonlinear relationships between earnings management and corporate tax avoidance in all five countries. Notably, their findings show that standard linear regression cannot capture these dynamics. This shows that it is very important to have machine learning (ML) models that can adapt when working with payroll and tax information, where the assumptions about the data are often not met. This methodological knowledge influences the selection of the forecasting program in Task 4 of this coursework.

Alexi Lazebnik and Shami(2024)[5] suggest a new way predicting how much tax a country will collect. Their model uses a computer program called a genetic algorithm and special type of computer program called an agent simulation. They demonstrate that models accounting for population heterogeneity outperform aggregate econometric approaches when examining data beyond the firm level. Their insight that sub-populations of taxpayers behave differently is directly applicable to the unsupervised clustering in task 5. Grouping employees by sector before modelling may reveal a latent structure that a single aggregate model would obscure. Together, these studies establish that machine learning (ML)-based approaches to tax data analysis are empirically superior to traditional methods, provided the underlying data quality is adequate. A condition that renders Tasks 2 and 3 of this coursework foundational rather than incidental.

Summarised papers in tables

Ref Number and Year of the paper	Fuest, C., Peichl, A. and Siegloch, S. (2018)
Country/Region	Germany
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Corporate income tax; wage incidence (Einkommensteuer / Lohnsteuer)
Machine Learning Algorithms Applied	Difference-in-differences (DiD), event study and panel regression.
What are the key findings or contributions of this paper	Workers account for around 50% of the corporate tax burden. Low-skilled, young and female employees take on an even larger share. The incidence effect is stronger in firms with collective bargaining agreements.
What insight did you gain out of the paper	It has been found that tax changes are not spread evenly through wages based on worker characteristics. This has highlighted the potential for structural bias in industry-level pay data in this coursework. It has been understood that different Lohnsteuer and PAYE responses will be shown by industries with different gender and skill compositions, and that these responses cannot be captured by a single regression line.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study only covers the local business tax (Gewerbesteuer), not the Lohnsteuer schedule. Additionally, the DiD method cannot easily be extended to predict future values, as required for Task 4. A supervised machine learning (ML) approach, trained on longer time-series payroll data, could complement these findings.

Table 1: Summarised findings of first paper [1], 2018

Ref Number and Year of the paper	Lahann, J., Scheid, M. and Fettke, P. (2019)
Country/Region	Germany / EU
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	VAT and payroll-related accounting data
Machine Learning Algorithms Applied	Decision trees, ML classification
What are the key findings or contributions of this paper	Decision tree classifiers can successfully identify VAT compliance violations in German accounting data. Tree-based rules can be understood by auditors and can be applied to new data.
What insight did you gain out of the paper	For the prediction task in this coursework, tree-based methods (including Random Forest and Gradient Boosting variants of decision trees) are strong candidates because the feature space includes continuous pay variables and categorical industry labels, which is a mixed structure that suits these algorithms.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study addresses binary detection (violation/no violation) and does not quantify the magnitude of the revenue lost through under-reported tax. Extending the study to a regression framework to estimate the scale of under-reported tax would substantially increase its operational usefulness for tax authorities, as this would be directly analogous to predicting mean pay values in Task 4.

Table 2: Summarised findings of second paper [2], 2019

Ref Number and Year of the paper	Delgado, F.J., Fernández-Rodríguez, E., García-Fernández, R., Landajo, M. and Martínez-Arias, A. (2023)
Country/Region	Germany, UK, France, Italy, Spain
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Corporate income tax / effective tax rate (ETR)
Machine Learning Algorithms Applied	Artificial neural networks (ANN), nonparametric regression
What are the key findings or contributions of this paper	ANN regressions reveal nonlinear relationships between earnings management and corporate tax avoidance. These results are consistent across all five major EU economies, including Germany. Standard linear models fail to detect these patterns.
What insight did you gain out of the paper	This paper justifies the use of artificial neural network (ANN)-style or nonlinear machine learning (ML) models over simple linear regression for Tasks 4 and 5 of this coursework. The principle that income and tax variables rarely conform to linear relationships is confirmed empirically at the EU scale.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The study period ends in 2015. This excludes the impact of post-2017 EU anti-tax avoidance directives, ELStAM digital reporting reforms and the shock to corporate tax data caused by the onset of the pandemic. Updating the ANN models with a decade of additional data would reveal whether the identified nonlinear relationships remain stable through structural breaks.

Table 3: Summarised findings of third paper [3], 2023

Ref Number and Year of the paper	Götttert, M. and Lehmann, R. (2021)
Country/Region	Germany
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Income tax and Lohnsteuer revenue forecasting
Machine Learning Algorithms Applied	Econometric error decomposition, regression analysis of forecast components
What are the key findings or contributions of this paper	Errors in tax revenue forecasts by Germany's official estimation working group are primarily driven by mis-estimation of the tax base, rather than errors in elasticity. This finding challenges the conventional focus on refining model parameters at the expense of data quality.
What insight did you gain out of the paper	This paper is the most directly applicable to this coursework. It explains why the corruption and missing data in the COMP1886 dataset cause forecasting problems, and why pre-processing in Task 3 is essential; poor data will produce poor predictions regardless of the ML algorithm applied in Task 4.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The paper does not test whether machine learning (ML)-based forecasting could outperform the traditional econometric methods used by the official forecasting group. Given the richness of ELStAM data, however, this comparison would be both feasible and highly relevant to policy.

Table 4: Summarised findings of fourth paper [4], 2021

Ref Number and Year of the paper	Alexi, A., Lazebnik, T. and Shami, L. (2024)
Country/Region	Cross-national / stylised economies
Type of Taxation Discussed (PAYE / Property / GST / Capital etc.)	Income tax revenue forecasting
Machine Learning Algorithms Applied	Genetic algorithm (GA), heterogeneous agent modelling, ML-based optimisation
What are the key findings or contributions of this paper	A microfounded general equilibrium (GA) model with heterogeneous taxpayer populations produces more accurate tax revenue forecasts than aggregate econometric models. It has been shown that modelling sub-population variation is critical for forecast quality.
What insight did you gain out of the paper	Both Germany's Lohnsteuerklassen system and the industry-sector breakdown in this coursework dataset encode the kind of taxpayer heterogeneity that is critical to capture, as shown in [5]. This directly motivates the clustering approach in Task 5, whereby employees are grouped by behavioural and structural similarity before modelling to reveal patterns that a single undifferentiated model would miss.
Additional Concerns that still exists in the paper which you think the Authors Could Have Addressed	The model has only been validated on stylised, simulated economies. Applying it to Germany's actual Lohnsteuer time series data, which is publicly available in aggregate form from Destatis, would provide crucial real-world validation of the heterogeneous agent approach and produce findings of immediate policy relevance.

Table 5: Summarised findings of fifth paper [5], 2024

References:

- [1] Fuest, C., Peichl, A. and Siegloch, S. (2018) 'Do higher corporate taxes reduce wages? Micro evidence from Germany', *American Economic Review*, 108(2), pp. 393–418. Available at: <https://www.aeaweb.org/articles?id=10.1257/aer.20130570> (Accessed: 24 March 2026).
- [2] Lahann, J., Scheid, M. and Fetteke, P. (2019) 'Utilizing machine learning techniques to reveal VAT compliance violations in accounting data', in *Proceedings of the 2019 IEEE 21st Conference on Business Informatics (CBI)*, Moscow, Russia, 9–11 July 2019. IEEE. Available at: <https://ieeexplore.ieee.org/abstract/document/8808015> (Accessed: 24 March 2026).
- [3] Delgado, F.J., Fernández-Rodríguez, E., García-Fernández, R., Landajo, M. and Martínez-Arias, A. (2023) 'Tax avoidance and earnings management: a neural network approach for the largest European economies', *Financial Innovation*, 9(1), pp. 1–25. Available at: <https://link.springer.com/article/10.1186/s40854-022-00424-8> (Accessed: 24 March 2026).
- [4] Göttert, M. and Lehmann, R. (2021) *Tax revenue forecast errors: wrong predictions of the tax base or the elasticity?* CESifo Working Paper No. 9148. Munich: CESifo. Available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3872387 (Accessed: 24 March 2026).
- [5] Alexi, A., Lazebnik, T. and Shami, L. (2024) 'Microfounded tax revenue forecast model with heterogeneous population and genetic algorithm approach', *Computational Economics*, 63(5), pp. 1705–1734. Available at: <https://link.springer.com/article/10.1007/s10614-023-10379-2> (Accessed: 24 March 2026).

TASK 2: Exploratory Data Analysis (EDA)

Hypothesis 1: Sector-Level Relationship between Employment and Mean Pay

This hypothesis examines whether industries with higher employment levels also have higher average pay. To investigate this, both given datasets were combined at the sector level, and Pearson correlations were calculated across the industries. The Pearson correlation coefficient(r) was used to measure the strength and direction of the linear relationship between the number of employees and average pay, across sectors and over time.

There was a strong positive correlation in fields such as health and social work ($r \approx 0.98$) and professional and scientific activities($r \approx 0.92$). This indicates that wages tend to increase alongside employment growth, likely driven by higher demand for skilled labour. By contrast, negative correlations were observed in sectors such as manufacturing($r \approx -0.51$) and wholesale and retail($r \approx -0.56$). This indicates that increased employment does not always lead to higher salaries.

These outcomes show/highlight that the relationship between employment and pay varies by sector, reflecting structural differences in productivity and skill composition. These findings suggest that a single global ML-model may oversimplify the data. These heterogeneous patterns can only be captured effectively using sector-specific features or nonlinear models.

Hypothesis 2: Macro-level co-movement of total employment and average pay

The second hypothesis explores whether increases in total employment are associated with rising average pay overtime. Combined monthly statistics demonstrate a notable positive correlation($r \approx 0.84$) between the two metrics, as confirmed by an increasing trend. However, short-term fluctuations suggest the influence of external factors, meaning aggregate trends may hide variations within sectors.

This suggests that although overall characteristics provide valuable predictive information, combining them with industry-specific data is essential for improving accuracy and avoiding aggregation bias.

Visualisations of EDA

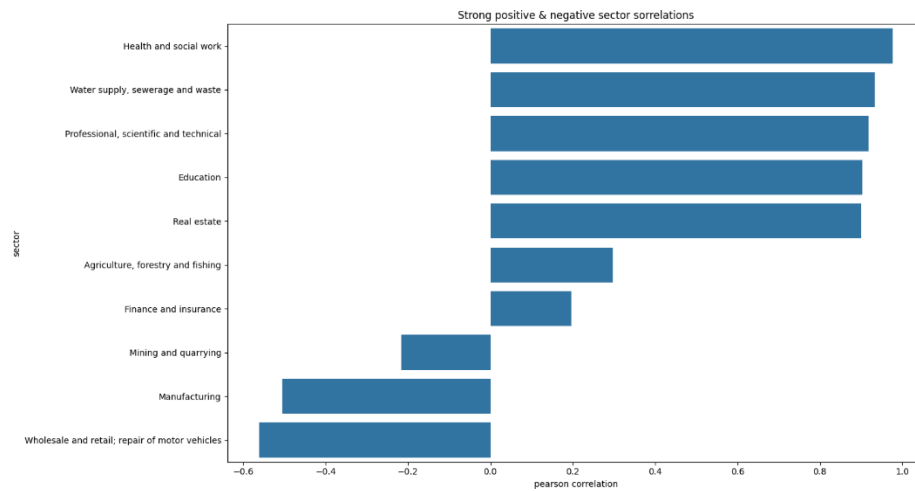


Figure 1: EDA Hypothesis 1 – horizontal bar chart of Pearson correlation between PAYE employee count and mean pay

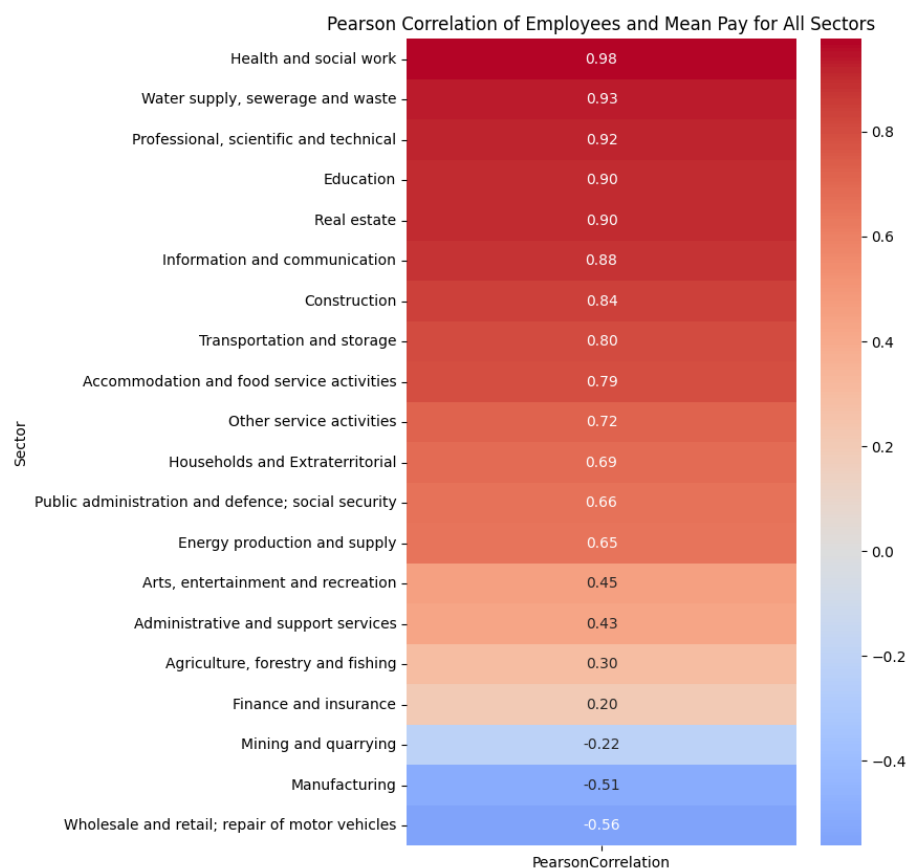


Figure 2: EDA Hypothesis 1: Heatmap Pearson correlation

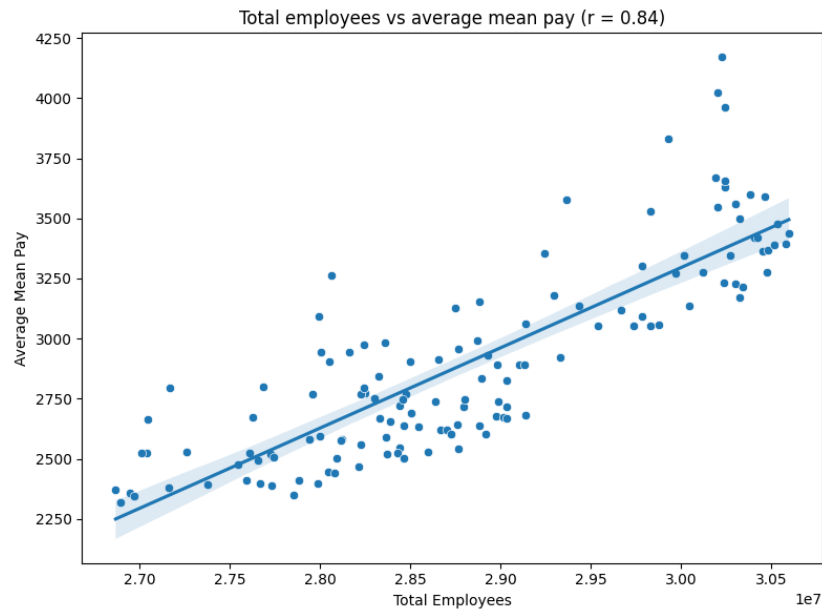


Figure 3: EDA Hypothesis 2 – scatter plot with OLS regression line

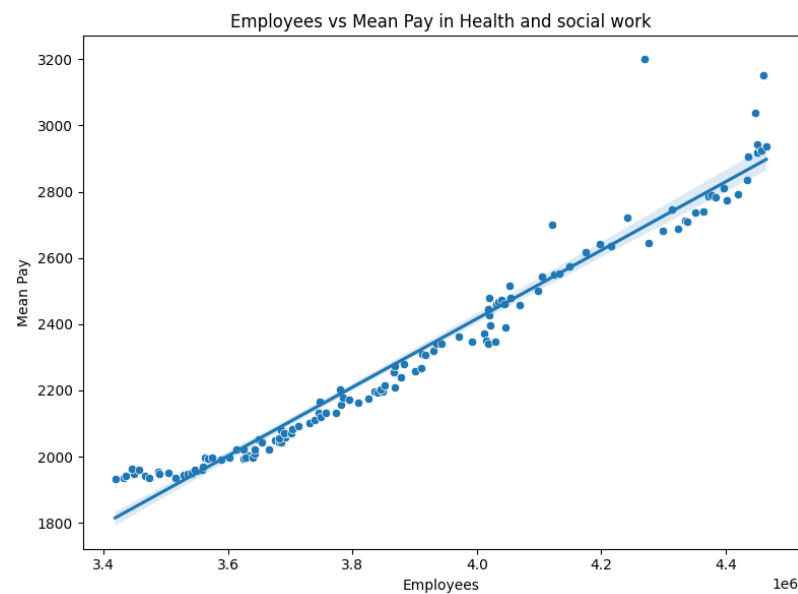


Figure 4: EDA Hypothesis 2 – scatter plot with OLS regression line 2

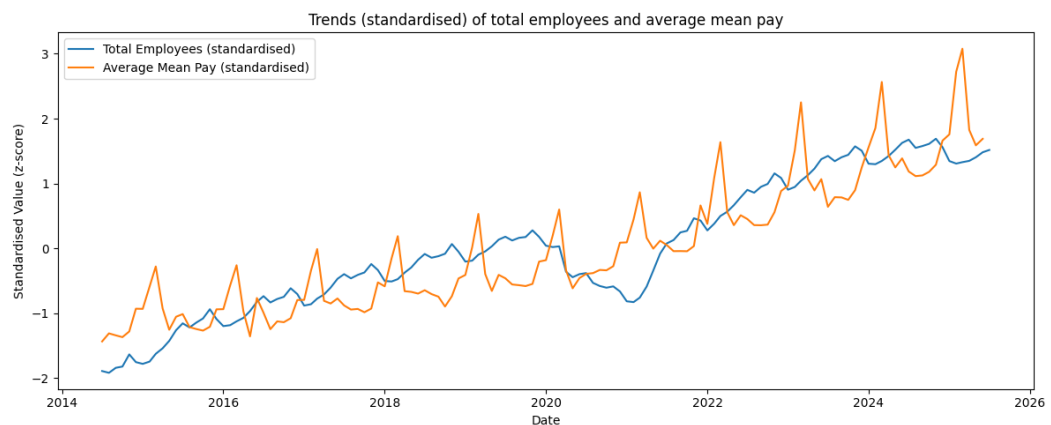


Figure 5: EDA Hypothesis 2 – dual standardised (z-score) time series of total PAYE employees and average pay

TASK 3: Data Pre-Processing

Technique 1: Missing Value Imputation

Missing values were identified in the 'Mean_Pay' dataset using visualisation and summary statistics. It reveals sporadic gaps across multiple sectors. Firstly, linear interpolation was used to estimate missing values. Leveraging the time-series structure by assuming gradual changes between consecutive months. This method was selected as wage data typically evolves smoothly rather than exhibiting abrupt fluctuations.

Remaining missing values, particularly at dataset boundaries where interpolation is not feasible, were replaced using sector-specific mean imputation. This ensured data completeness while minimising distortion of the overall distribution. However, mean imputation may reduce variability and introduce bias if missing values are not randomly distributed.

Alternative methods were not used: row deletion would reduce dataset size and remove valuable temporal information, while forward filling could introduce bias by artificially smoothing trends.

Technique 2: Data Transformation and Feature Preparation

For the second technique, the column 'Date' was converted into a datetime format and set an index to enable a time-series analysis. Additionally, all sector columns were converted to numeric format. Invalid values have been replaced with NaN prior to imputation, i.e., zeros showing missing data.

Both datasets were aligned by merging the common attributes. This ensures consistency between the input feature of employment and the target variable of mean_pay. Also, StandardScaler was used to normalise the features, which is important for K-means clustering in Task 5. Because Euclidean distance is sensitive to scale.

These steps improve the quality of the data, keep the temporal structure and let machine learning models find meaningful patterns. This directly improves how well Task 4 can predict.

Visualisations & Findings of Data Pre-processing

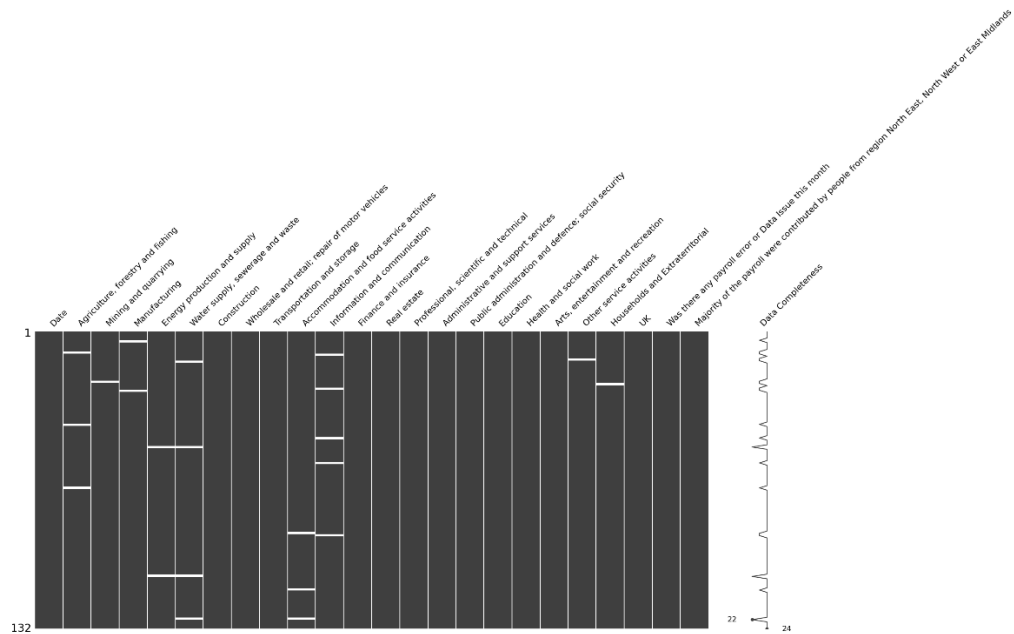


Figure 6: Missing Values in Dataset 'Mean Pay'

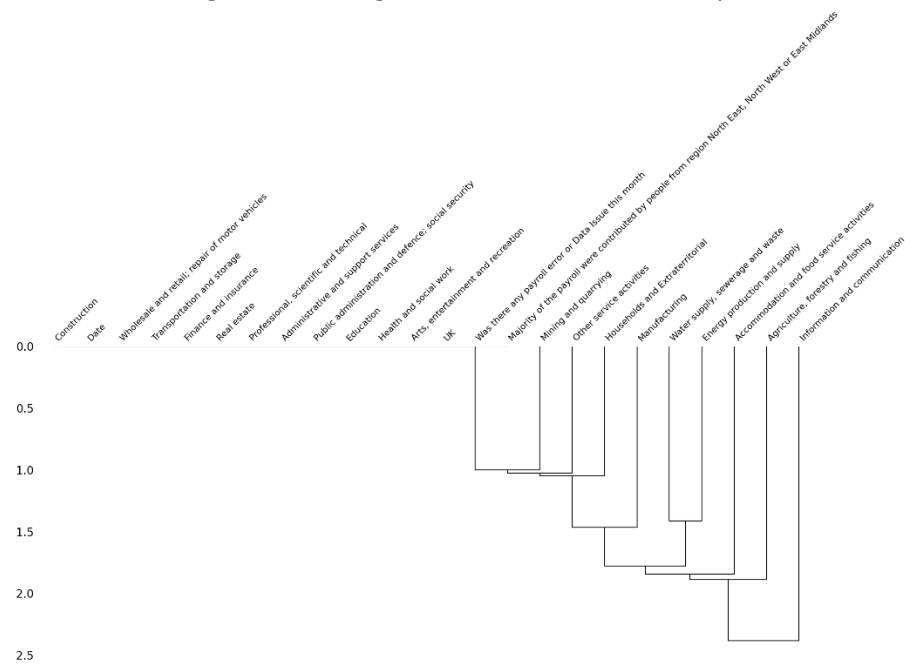


Figure 7: Missing Values Illustration 2

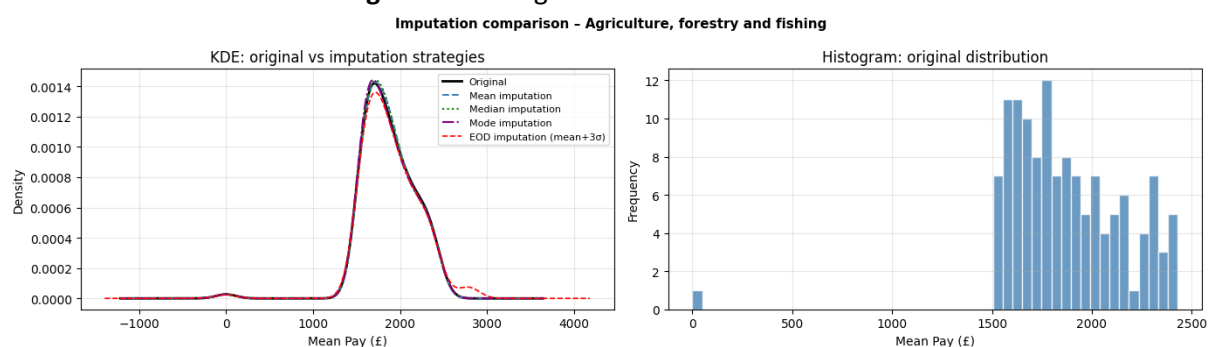


Figure 8: KDE comparison for sector: 'Agriculture, forestry and fishing' using the original, mean, median and mode strategies, as well as EOD imputation.

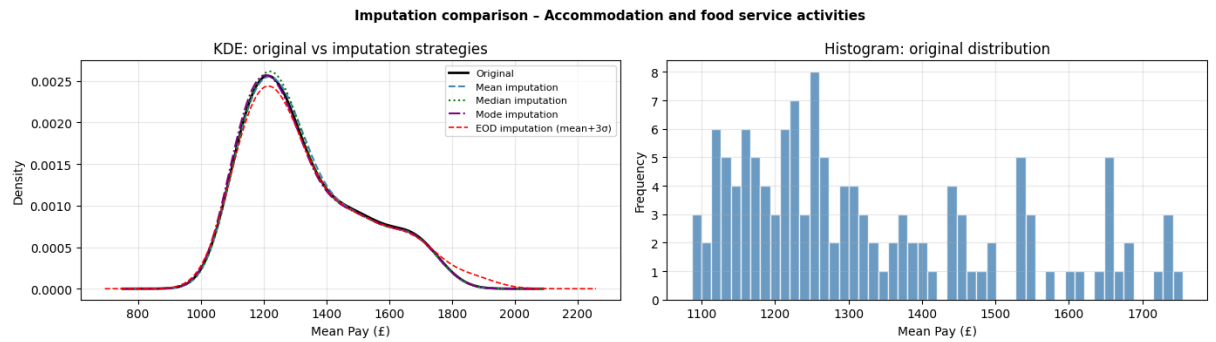


Figure 9: KDE comparison for sector: Accommodation and food service activities

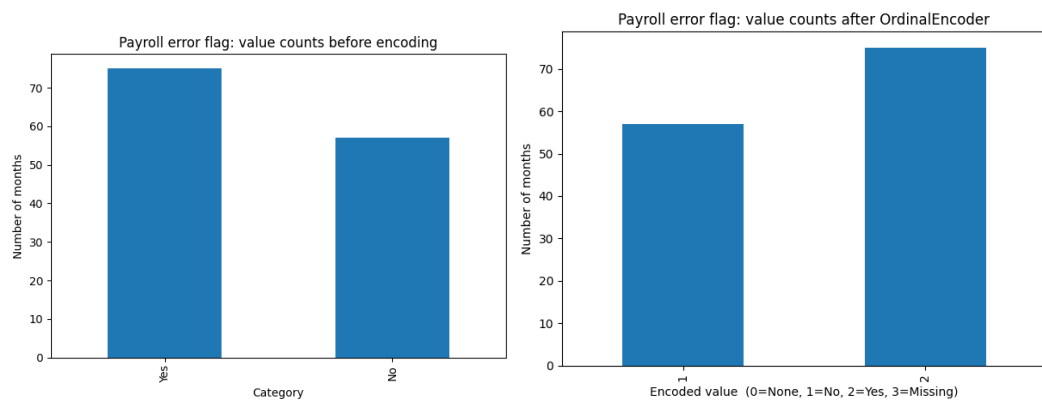


Figure 9: Payroll error flag value counts before and after OrdinalEncoder (1=No, 2=Yes).

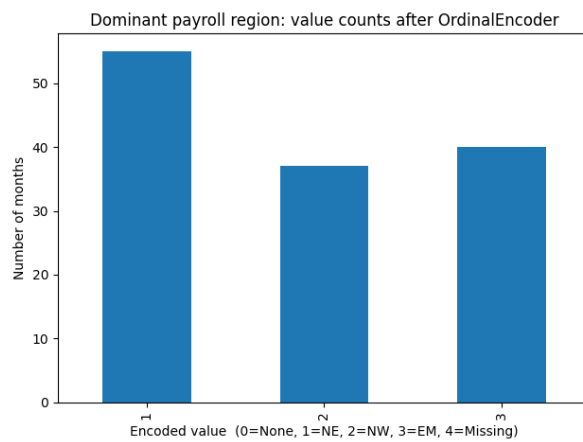


Figure 10: dominant payroll region

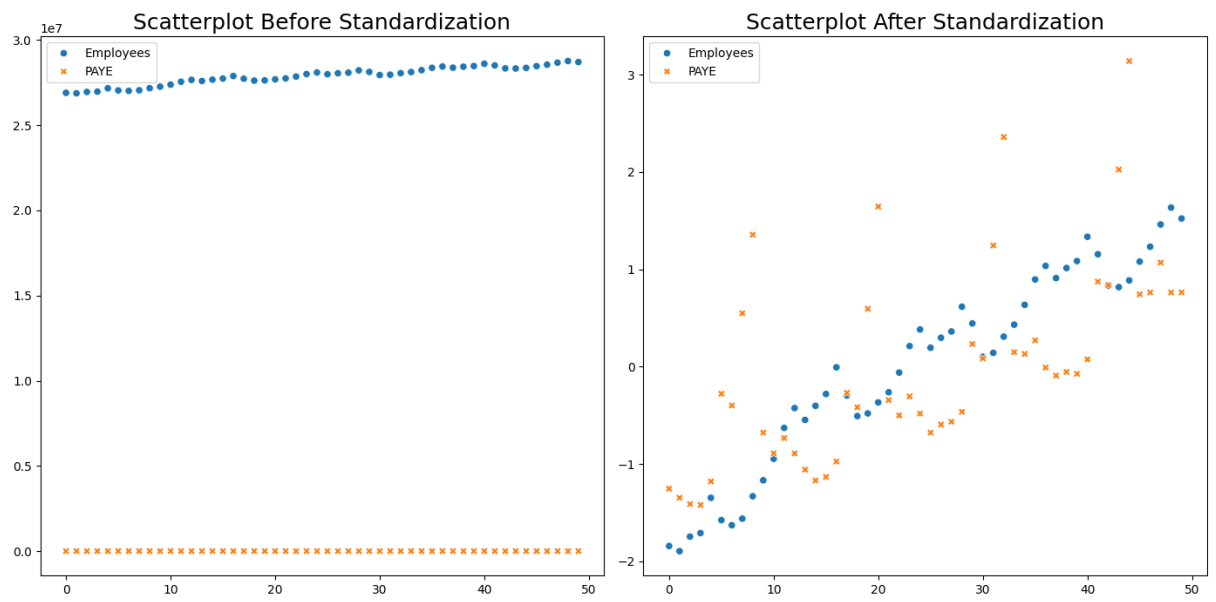


Figure 11: Scatterplots before and after StandardScaler applied to Employees and PAYE columns.

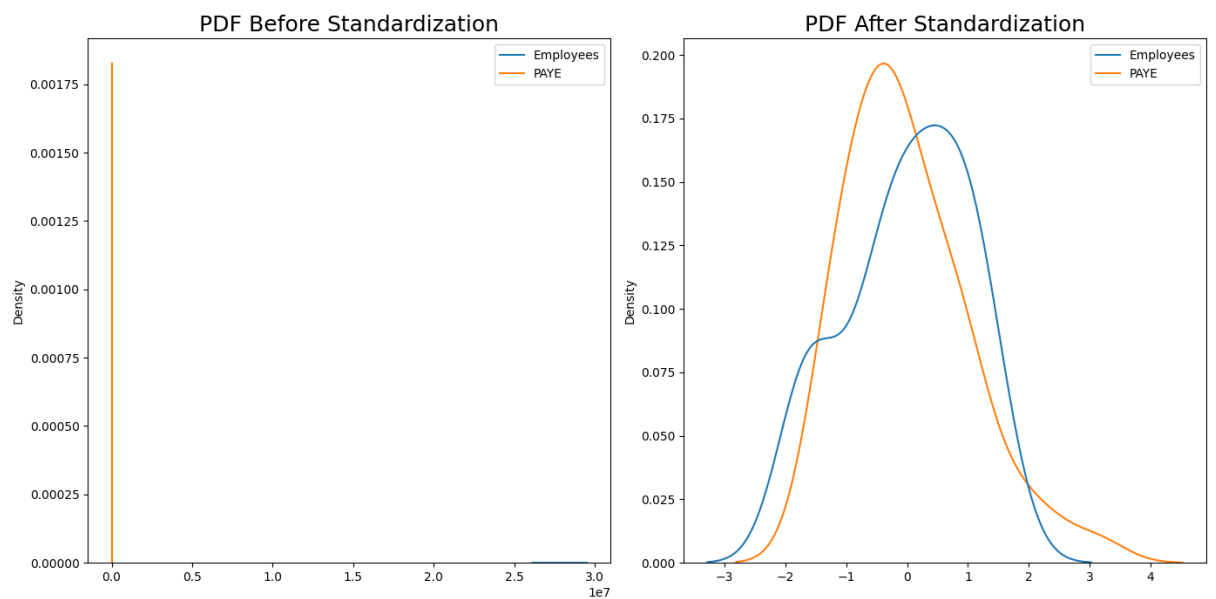


Figure 12: KDE distributions before and after StandardScaler, confirming mean = 0, std = 1.

TASK 4: Supervised ML algorithm

Summary

Two supervised machine learning models were used to predict the mean pay for June 2025: Linear Regression and Random Forest Regressor. Linear Regression is a parametric model that assumes a linear relationship between the input features and the target variable. It then finds coefficients that minimise the error in predictions. It is highly useful due to its simplicity and ease of interpretation. Random Forest is an ensemble learning method that builds several decision trees from random subsets of data and features. The final predictions are made by averaging the results. This enables the model to capture nonlinear relationships and interactions between variables

This model choice allows a comparison between a straightforward linear approach and a more adaptable nonlinear model. Results from Task 2 already revealed significant variability in sector-level relationships. This indicates that a global linear assumption might be inadequate. Therefore, Random Forest would be expected to perform better, as it can capture more complex patterns across sectors.

The dataset was prepared for modelling through preprocessing. By cleaning the Mean Pay dataset by encoding and removing unnecessary categorical columns, the sectors became individual observations. To match up the input features and target variables, both datasets were combined. A time-based split was used for the train and test sets. All observations before June 2025 were used for training, and June 2025 was used as the test set. This method prevents data leakage and provides a realistic way to make predictions.

Several metrics were used to evaluate the model's performance, including Mean Squared Error(MSE), Root Mean Squared Error(RMSE), Mean Absolute Error(MAE), and R^2 score. This gives a full picture of how accurate the predictions were and their explanatory power.

The results show that Linear Regression outperforms the Random Forest model. It achieves lower error values and a significantly higher R^2 score. This indicates that Linear Regression explains a larger proportion of the variance in mean pay for June 2025. This is further supported by visual analysis, where Linear Regression predictions align more closely with the ideal diagonal line. While Random Forest shows larger deviations, particularly in sectors with more complex wage dynamics.

The dataset characteristics help explain the performance difference. While Task 2 indicated variability across sectors, the strong performance of Linear Regression suggests that, at this level of aggregation, the overall relationship between employment and pay remains largely linear. As a result, the linearity assumption does not significantly limit model performance. In contrast, Random Forest may be less effective due to the relatively small dataset, which limits its ability to learn complex nonlinear patterns.

Both models perform relatively well in sectors with stable trends, such as health and social work. However, larger prediction errors are observed in sectors such as manufacturing and wholesale and retail, where wage dynamics are influenced by external factors not included in

the dataset. This highlights a key limitation: model performance is constrained by the available features.

Overall, the results show that choosing the right model has a significant effect on how accurate predictions are. Linear Regression proves to be the more suitable model for this task, as it effectively captures the dominant linear relationship in the data. While Random Forest offers greater flexibility, its performance is limited in this case. This highlights that model performance depends not only on model complexity but also on the underlying structure of the data.

Future improvements may involve the integration of additional features, such as inflation or GDP indicators, alongside the utilisation of time-series-specific models (e.g., ARIMA or LSTM) to more effectively capture temporal dependencies.

Metric	Linear Regression	Random Forest
MSE	29870	1293366.25
RMSE	172.83	1137.26
MAE	151	828.25
R ²	0.9865	0.4151

Table 6: Metrics of supervised ML methods

Visualisations supervised ML

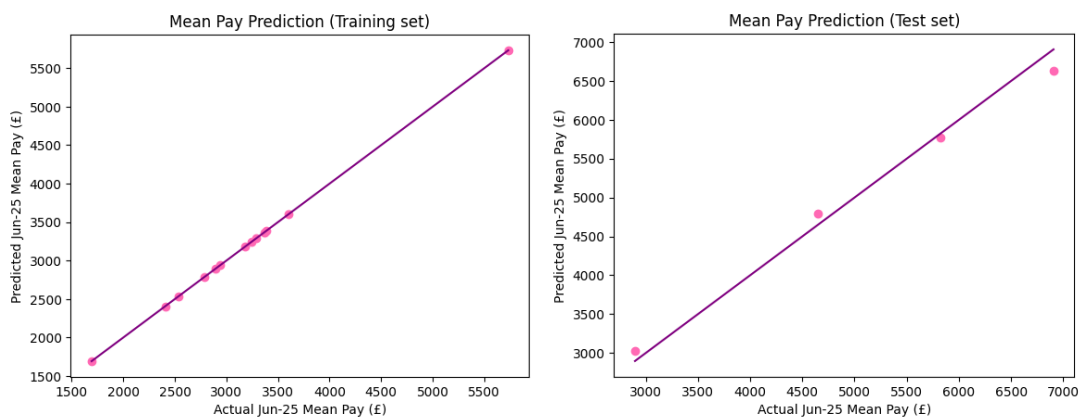


Figure 13: Linear Regression – training & test set

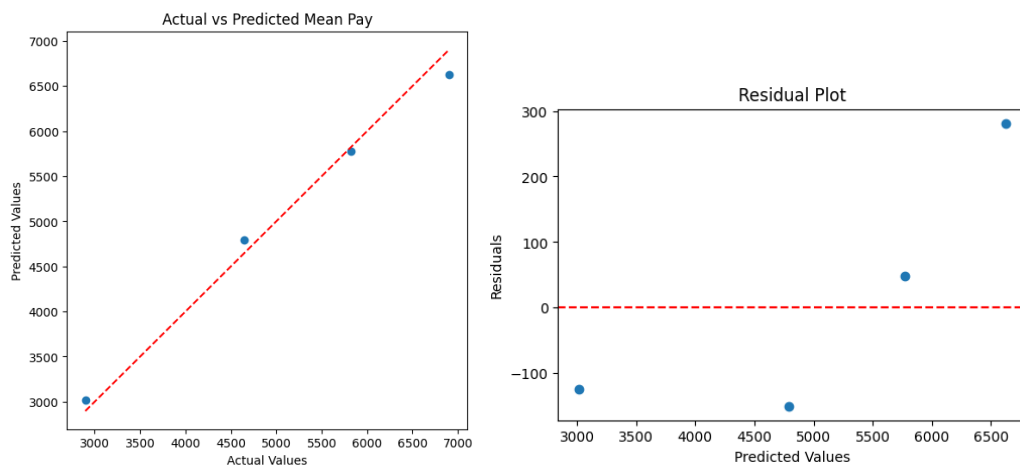


Figure 14: Linear regression – actual vs predicted mean pay & residual plot

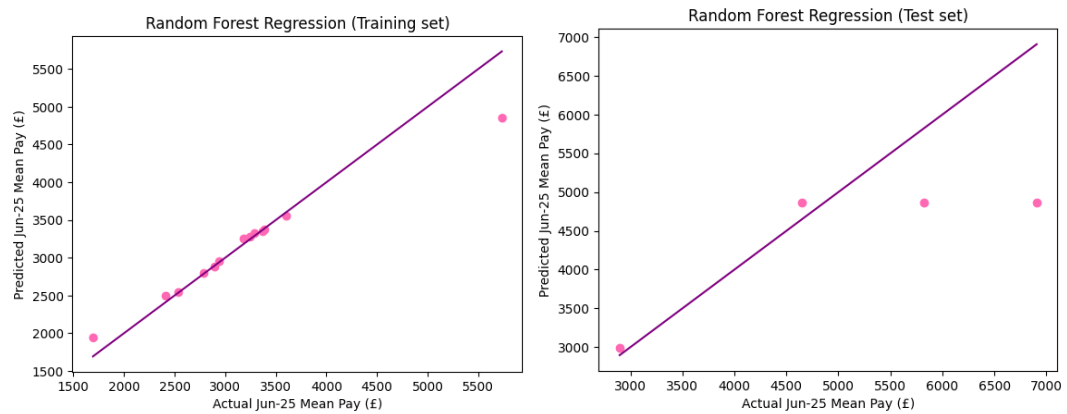


Figure 15: Random forest – training & test set

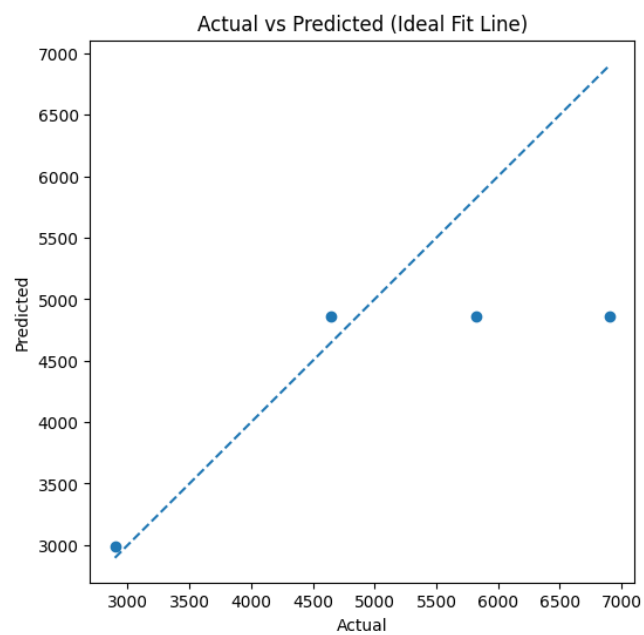


Figure 16: random forest – actual vs. predicted

TASK5: Unsupervised ML algorithm

Summary

K-Means clustering has been applied to the first 50 observations of the dataset to examine the relationship between Employees and PAYE. K-Means was selected because it is fast and works well with numerical data. It divides observations into k clusters by minimising the within-cluster variance using Euclidean distance. Distance-based clustering is appropriate because both are continuous variables. Before clustering, StandardScaler was used to standardise the features so that differences in scale did not impact the distance calculations.

Whereby, hierarchical methods offer a structured view of cluster relationships and DBSCAN can identify non-spherical and noisy clusters. K-means was selected because of its computational efficiency, scalability and interpretability.

Both the elbow method and silhouette analysis were used to find the optimal number of clusters. The elbow plot clearly shows an inflection point at $k = 4$, where the drop in the within-cluster sum of squares starts to level off. The silhouette score supports this, reaching its maximum (around 0.4434) at $k = 4$. This makes the selection stronger than methods relying on guesswork.

The resulting clusters reveal distinct employment-pay structures. One cluster represents low-pay, moderate-employment sectors, while other captures high-pay, high-employment sectors, reflecting differences in productivity and skill intensity across industries. Intermediate clusters illustrate gradual transitions between these extremes. However, the clusters exhibit partial overlap, particularly in mid-range values, suggesting that the relationship between Employees and PAYE is continuous rather than strictly segmented.

The moderate silhouette score indicates limited cluster separation, highlighting a limitation of K-means clustering, which assumes clusters are well-separated and spherical. This suggests that alternative methods, such as DBSCAN or Gaussian mixture models, may be more effective in capturing overlapping or complex structures. These findings are consistent with those of Alexi et al. (2024), who demonstrate that economic sub-populations frequently do not form sharply defined clusters.

Visualisations of unsupervised ML

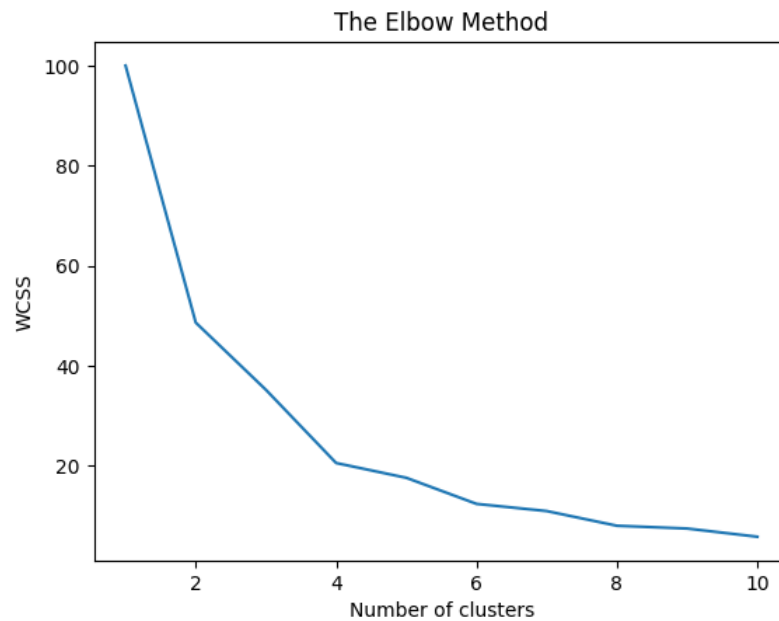


Figure 17: Elbow Method plot: WCSS against number of clusters ($k = 1$ to 10), used alongside the Silhouette Score to identify the optimal k .

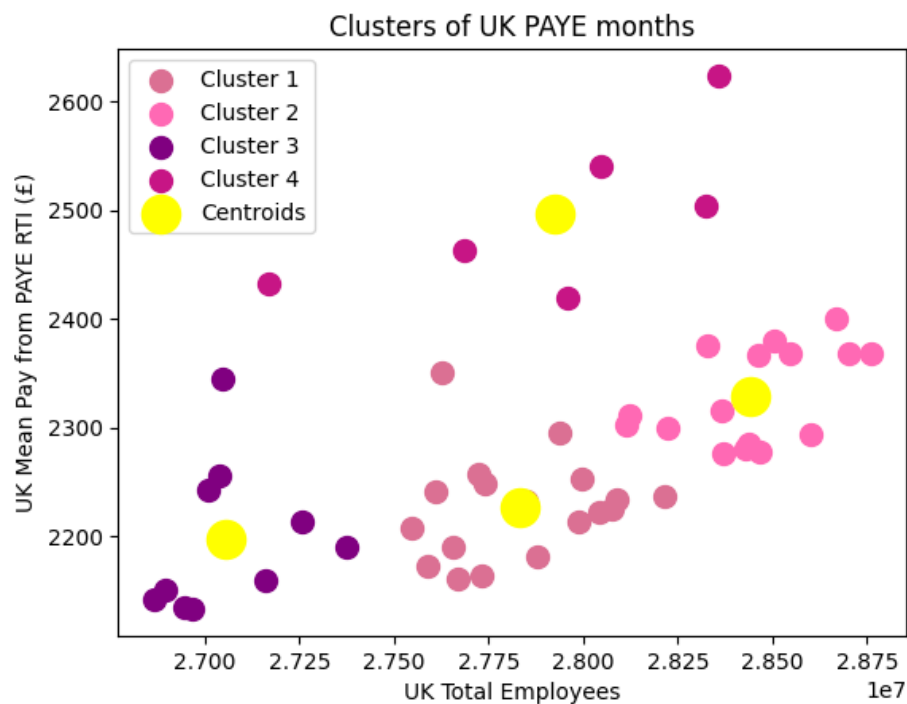


Figure 18: K-Means cluster scatter plot of UK total employees vs UK mean PAYE pay for the first 50 tuples, with cluster centroids marked in yellow.

TASK 6: Conclusion

This project demonstrated a full-cycle machine learning pipeline, combining exploratory data analysis with predictive modelling to examine the relationship between employment and PAYE across sectors.

The EDA revealed meaningful patterns in sectoral variation and aggregate relationships. These insights were effectively carried forward into the modelling stage. Both Linear Regression and Random Forest were implemented. Linear Regression demonstrated the strongest predictive performance, achieving a significantly higher R^2 score than Random Forest. This indicates that the relationship between employment and PAYE is largely linear at this level of aggregation.

The clustering analysis enhanced the study by uncovering underlying group structures within the data. The use of both the Elbow Method and Silhouette Score supported the selection of four clusters, strengthening the reliability of the results. However, partial overlap between clusters suggests that the relationship is continuous rather than strictly segmented.

Future improvements could include incorporating additional features such as macroeconomic indicators (e.g., inflation, GDP), as well as applying time-series models to better capture temporal dependencies. During implementation, minor challenges such as data formatting and alignment were encountered (i.e., merging datasets and preparing inputs). These were successfully resolved through careful preprocessing and validation. This showcases the importance of data preparation in ML workflows.

In my opinion, I found Task 4 most interesting because Linear Regression outperformed Random Forest, which contradicted the initial expectation from Task 2 that more complex, nonlinear models would perform better. This highlights that model performance depends not only on model complexity but also on the underlying structure of the data, which in this case appears to be predominantly linear. Furthermore, Task 5 is intriguing, as K-means clustering revealed hidden temporal structure in the unlabelled data, showing how unsupervised methods can generate hypotheses for further analysis. Overall, the project highlights the effectiveness of combining data analysis with machine learning techniques to generate meaningful insights and reliable predictions.

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